

Composition Pair: Three Adjacencies × AI-as-Substrate-Mediator — AI Mediation Across Adjacencies as the Architectural Pattern Where Mediator Role Operates at Adjacency Boundaries in the Coordination Knowledge Substrate Pattern

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This work derives from and formalizes the Coordination Knowledge Substrate (CKS) pattern introduced in *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026). It does not introduce new axioms. Its sole contribution is to formalize the composition of two foundational architectural commitments — three adjacencies (CKS is not RAG, not parametric memory, not external structured memory) and AI-as-substrate-mediator — into the emergent architectural property neither commitment yields alone: AI mediation across adjacencies, the pattern under which the mediator role operates at adjacency boundaries when CKS substrates compose with adjacent AI components.

Abstract

The CKS pattern’s three-adjacencies commitment names the architectural distinction between CKS and the three adjacent design objects with which it is most often conflated. The AI-as-substrate-mediator commitment names the architectural role LLMs play in CKS systems through five properties: substrate-content reads as primary state, substrate-content writes under orchestration rules, no substrate-relevant state outside the substrate, no authority over substrate content, and LLM writes recorded as substrate content with attribution. Each commitment has been formalized as a standalone derivation in this series. This note formalizes the *composition* of the two — what becomes architecturally specified when both commitments hold together at the boundary where a CKS cell consults an adjacent AI component. The composition produces an emergent architectural property, *AI mediation across adjacencies*, where the mediator role operates at adjacency boundaries with mediator properties preserved on both sides of any cell-to-adjacent-component consultation. The note articulates the property as four operational components, identifies what the composition forces beyond either commitment in isolation, names the anti-patterns specifically violating it, distinguishes it from broader mediator-preservation and from consultation mechanics without mediator specification, and provides a three-property operational test.

1. Why the composition pair needs to be formalized as standalone

The three-adjacencies commitment specifies architectural positions adjacent design objects occupy relative to a CKS substrate but, in isolation, says nothing about how AI behaves at the boundary between a CKS cell and any of those adjacent positions. The AI-as-substrate-mediator commitment specifies the role LLMs play within a CKS cell over a CKS substrate but, in isolation, says nothing about how that role extends to AI operating at, across, or within an adjacent component. Each commitment is operationally

complete on its own scope; neither addresses what happens at the architectural seam where the two scopes meet.

That seam is where most real CKS deployments live. A regulated coordination workflow with a CKS substrate at its center is typically embedded in a larger AI system that includes a RAG retrieval index over reference documents, a fine-tuned model specialized to a domain vocabulary, a vector database the organization already operates, or an external structured memory store the substrate inherits from. Cell consultations of these adjacent components — the canonical case in which a cell uses an adjacent component as input to its execution, the case this note is scoped to — are where the architectural seam is actually traversed. If the mediator role is unspecified at that traversal, the cell's writes back to the substrate may carry no provenance for what the adjacent component contributed, the adjacent component's AI may exercise authority over what the cell writes, and cross-adjacency operations may compound mediator failures invisibly across multiple boundary crossings. The standalone formalization makes the seam architecturally addressable: it states what the cell-side AI must satisfy, what the adjacency-side AI must satisfy, what the boundary itself must preserve, and what cross-adjacency operations must satisfy at each crossing.

This is the third note in the three-adjacencies composition cluster within Phase A4. The first formalizes the mapping between the three adjacencies and the three composition patterns by which adjacent components are positioned relative to the substrate; the second formalizes the adjacency-vs-substrate authority distinction; the present note formalizes the mediator role at adjacency boundaries; the next will close the cluster with the composition-requirements composition.

2. The emergent architectural property

The composition pair produces an emergent architectural property — *AI mediation across adjacencies* — that has four operational components. Each states a requirement the composition imposes that neither commitment, in isolation, imposes by itself.

Component 1 — Cell-side mediator preservation. When a CKS cell consults an adjacent component during the cell's execution, the LLM operating inside the cell continues to operate as substrate mediator throughout the consultation. All five properties hold for the cell-side LLM during the consultation phase as they hold during the cell's reasoning over substrate content alone: substrate content remains the primary source of state; writes back to the substrate remain under the orchestration rule that authorizes the cell's execution; no substrate-relevant state is held outside the substrate as a result of the consultation; no authority over substrate content is exercised; and writes are recorded as substrate content with attribution including the consultation itself. The consultation is part of the cell's reasoning environment; it does not promote the cell-side LLM to a role beyond mediator.

Component 2 — Adjacency-side AI as mediator within the adjacency's architectural position.

When the adjacent component itself includes AI — a retrieval LLM internal to a RAG index, a fine-tuned model that responds to the cell's query, an LLM internal to an external structured memory store — that adjacency-side AI operates as mediator within the adjacency's own architectural position. It does not exercise authority over CKS substrate content (it cannot, because the consultation pattern holds the adjacency non-authoritative on coordination questions), does not hold CKS-substrate-relevant state

outside the CKS substrate, and does not produce outputs that flow into the CKS substrate without passing through the cell's orchestration rule. Where the adjacent component is a vendor product whose internal architecture is partially opaque, the architectural test is whether its outputs *can* be received by the cell as inputs to mediator-governed reasoning, not whether the vendor's internal LLM operations are themselves CKS-mediator-compliant in their own scope.

Component 3 — Mediator properties span the adjacency boundary through the consultation pattern. The cell-to-adjacent-component consultation is the operational mechanism by which mediator properties span the boundary. The cell reads the adjacent component as input; the adjacent component returns content; the cell reasons over that content under its orchestration rule; the cell writes back to the substrate with provenance including the consultation. At each phase, mediator properties hold on the side that owns the operation. The composition states that the consultation pattern's three boundary requirements — provenance crosses cleanly, substrate is authoritative on coordination questions, cell outputs remain orchestration-rule-governed — are mediator-role consequences, not separate boundary axioms.

Component 4 — Cross-adjacency AI operations preserve mediator at each crossing. When a cell's reasoning draws on more than one adjacent component — a cell that consults a RAG index and a fine-tuned model in the same execution, a cell whose RAG consultation passes through an intermediate external structured memory store before reaching the source corpus — mediator properties hold uniformly at each boundary crossing. Each crossing carries the cell-side mediator role into a new boundary interaction, and each must satisfy the same requirements component 3 specifies. Operations that compound a mediator failure invisibly across crossings fall outside the composition.

These four components are the operational content of *AI mediation across adjacencies*.

3. What the composition forces beyond either commitment alone

Three architectural consequences follow from the composition that do not follow from either commitment in isolation.

Properties A–E hold at adjacency boundaries, not only within cells. The mediator commitment alone scopes the five properties to LLM operations the cell performs over the substrate. The composition extends the scope to cell-side LLM operations that traverse a consultation boundary, with the consultation phase treated as part of the cell's reasoning environment. The cell's primary state remains substrate content even when adjacent input enriches its context; writes back to the substrate remain under the orchestration rule even when they incorporate adjacent-sourced content; reasoning over adjacent input does not produce substrate-relevant state held outside the substrate; adjacent components do not acquire authority by being consulted; and writes record the consultation with attribution.

Consultations are mediator-on-both-sides operations. The consultation pattern in isolation states three boundary requirements without naming the mediator role on the adjacency side. The composition states that the adjacency side has a mediator role within its own architectural position, and that this is what makes the boundary requirements coherent rather than asymmetric. A boundary across which only one side preserves a mediator role and the other is unspecified is not a boundary the composition covers.

Cross-adjacency AI is architecturally specified, not implementationally hand-waved. Neither commitment in isolation addresses configurations in which a single cell execution touches more than one adjacent component. The composition states that cross-adjacency operations preserve mediator at each crossing, that each crossing is independently subject to the boundary requirements, and that compounding mediator failures across crossings is itself an anti-pattern. Vendor adjacencies whose internal architecture is partially opaque are subject to the composition through architectural mapping: the deployment must verify that vendor outputs can be received as mediator-governed inputs even when the vendor’s internal LLM operations cannot be inspected.

4. Anti-patterns specifically violating the composition

Five anti-patterns name failures specifically of the composition pair. The first three are instantiations of canonical anti-patterns formalized elsewhere in this series; the last two are composition-specific.

Autonomous AI in adjacencies. The deployment hosts an autonomous LLM agent inside an adjacent component — a RAG-side agent that interprets the cell’s query and exercises judgment about what to retrieve and how to respond, a fine-tuned-model-side agent that produces multi-step outputs intended to drive the cell’s subsequent behavior. The autonomous agent holds substrate-relevant state outside the CKS substrate, exercises judgment over what the cell receives, and effectively operates as the authoritative source for content flowing into the cell’s reasoning. The fix is to bring the adjacency back to mediator-within-its-position: the adjacent component’s AI must respond to the cell’s query without holding cross-invocation state and without exercising judgment beyond what its architectural role authorizes.

Terminal-producer outputs across the boundary. The adjacent component’s outputs are treated as terminal artifacts the cell consumes whole — incorporated into substrate writes without rule-mediated interpretation, with the orchestration rule reduced to a passthrough. Property B fails because the cell’s writes are determined by the adjacent component’s outputs rather than by the rule operating over them; Property E fails because the writes’ attribution may name the cell while their effective authorship is the adjacency. The fix is to keep the rule operationally active: the cell’s reasoning over adjacent-sourced content must be a real interpretive operation under the rule.

Adjacency authority exercise. An adjacent component is treated as authoritative on a coordination question the substrate is the source of truth for — the cell defers to a RAG index on what was decided, accepts a fine-tuned model’s output as the answer to “what is the case,” queries an external structured memory store for substrate-relevant state and uses the response without going to the substrate. Property A fails because substrate content is no longer the primary source of state; Property D fails because the adjacent component is exercising authority over substrate content. The fix follows the adjacency-vs-substrate distinction: the adjacent component is consulted but not authoritative on coordination questions.

Cross-adjacency AI without mediator at every crossing. A cell’s reasoning passes through more than one adjacent component and mediator properties hold at the first crossing but degrade at subsequent crossings — a RAG consultation whose results pass through an intermediate summarization model not itself mediator-positioned, a fine-tuned model whose output is reformatted by a vendor service whose architectural role is unspecified. Each crossing must satisfy the boundary requirements independently; mediator

preservation does not transit through unspecified intermediate layers.

Vendor AI without mediator-mapping. The deployment includes a vendor adjacency whose internal AI architecture is opaque, and the deployment treats opacity as a deferral of the architectural question. The architectural test is whether the vendor’s outputs can be received by the cell as inputs to mediator-governed reasoning. A vendor whose outputs arrive with implicit authority claims, with substrate-relevant state in vendor-specific encodings, or with rule-bypassing side effects does not fit the composition; the deployment must either replace the vendor adjacency or reposition it as a non-consultation integration (a derived view of substrate content, a separate concern unrelated to coordination state, or out of architecture entirely).

5. What the composition is NOT

Stating what the composition is *not* is what keeps the formalization from drifting into something stronger than the source paper supports.

Not the three-adjacencies commitment alone. The commitment in isolation specifies architectural positions without specifying AI behavior at the boundaries those positions imply. A system that names its adjacencies cleanly but leaves AI behavior at boundaries unspecified satisfies the commitment without satisfying the composition.

Not the AI-as-substrate-mediator commitment alone. The commitment in isolation specifies five properties for LLM operations within a CKS cell over a CKS substrate without specifying boundary operations. A system that implements mediator-compliant cells but does not address what happens when those cells consult adjacent components satisfies the commitment without satisfying the composition.

Not the broader mediator-preservation composition. A separate composition pair (composition requirements $\times$ AI-as-substrate-mediator) covers preservation of the mediator role across the broader composition-requirements scope. The present note covers the mediator role specifically at adjacency boundaries within consultations. The two are operationally distinct: the broader pair is a composition-level preservation guarantee; the present note is the boundary-specific operational specification. A system can satisfy one while failing the other.

Not consultation mechanics without mediator specification. The consultation pattern as a mechanism names three boundary requirements; the present composition specifies the mediator role on both sides of the boundary as the deeper architectural commitment from which those requirements follow. The pattern without the mediator-on-both-sides specification is structurally a consultation but is missing the architectural justification for why the boundary requirements take the form they do.

6. Why the composition is load-bearing

The composition continues the three-adjacencies cluster within Phase A4 — the cluster that makes the three-adjacencies commitment operationally complete by specifying its mapping to the three composition patterns, its distinction from substrate authority, its mediator-role implications at boundaries, and its composition-requirements implications. It supports authority preservation at adjacency boundaries: adjacencies are non-authoritative on coordination questions per the adjacency-vs-substrate distinction; the op-

erational reason adjacencies remain non-authoritative is that AI at and across their boundaries operates as mediator per the present composition. It supports retraceability across boundaries: path retraceability does not stop at the cell boundary, and the mediator role at the boundary is what makes the trace traversable — writes carry attribution including the consultation, the consultation’s adjacency-side AI does not introduce trace breaks, and cross-adjacency operations preserve trace at each crossing. And it distinguishes CKS from boundary-breaking AI architectures: a system that hosts autonomous agents inside RAG components, treats fine-tuned-model outputs as terminal artifacts the cell relabels, lets external structured memory stores answer authoritatively on coordination questions, or compounds mediator failures across multi-hop AI operations may be a coherent system in some other architectural sense, but is not CKS-coherent.

7. Operational test

A system instantiates the composition — AI mediation across adjacencies — if and only if all of the following hold for every cell-to-adjacent-component consultation in the system, at all times during the system’s existence:

1. **(e.1) Pattern-A-mediator-both-sides.** For every consultation, the mediator commitment’s five properties hold for the cell-side LLM throughout the consultation, and the adjacency-side AI (if present) operates as mediator within the adjacency’s own architectural position — exercising no authority over substrate content, holding no substrate-relevant state outside the CKS substrate, and producing outputs the cell receives as inputs to rule-governed reasoning rather than as terminal artifacts.
2. **(e.2) Cross-adjacency-AI-mediator-uniform.** For every consultation that traverses more than one adjacent component, mediator properties hold at each boundary crossing independently. Compounding mediator failures across crossings — preservation at the first crossing followed by degradation at subsequent crossings — fails the test, regardless of how robustly the first-crossing properties hold.
3. **(e.3) Adjacency-AI-architecture-verified.** For every adjacent component the deployment uses under the consultation pattern, the deployment can verify (through documentation, vendor specification, architectural review, or direct inspection where possible) that the component’s AI architecture is compatible with mediator-role mapping — that its outputs can be received by the cell as inputs to mediator-governed reasoning. Vendor adjacencies whose architecture cannot be so verified must be repositioned (as a derived view, as a separate concern, or out of architecture) rather than left under the consultation pattern with the architectural question deferred.

A system that fails (e.1) at any consultation, fails (e.2) at any cross-adjacency operation, or fails (e.3) for any adjacent component does not implement the composition in the CKS sense. Such a system may be governed in some other architectural sense and may be useful for other purposes, but is not CKS-coherent on the adjacency-boundary axis.

One-sentence test. If every consultation in the deployment preserves the five mediator properties on the cell side and operates against an adjacency whose AI is mediator-compatible on its own side, with mediator properties uniformly preserved at every boundary crossing, the deployment instantiates AI mediation across adjacencies; if any consultation breaks any of these conditions, it does not.

8. Why naming the composition as standalone matters

The composition pair is not a derived consequence that downstream work will reconstruct from the two foundational commitments on its own. It is a place where two commitments meet at an architectural seam, and the operational content at that seam is not obvious from either commitment in isolation. Implementations that hold each commitment correctly in its own scope routinely violate the composition at the seam, because the seam is where the two scopes are silent and the deployment’s design fills the silence with whatever fits the moment’s engineering convenience.

Naming the composition as standalone gives downstream implementers a precise specification of what the seam requires. It makes the failure modes — autonomous AI in adjacencies, terminal-producer outputs, adjacency authority exercise, cross-adjacency mediator failures, vendor mediator-mapping omissions — explicitly identifiable rather than diffusely present, and positions the next note in the cluster on a foundation where the boundary-specific composition is already operationally pinned down.

Subsequent work that adopts the CKS pattern and uses one or more of the three adjacencies in hybrid configurations should use *AI mediation across adjacencies* in the sense formalized here. Subsequent work that hosts autonomous AI inside adjacencies, treats adjacency outputs as terminal artifacts the cell relabels, lets adjacencies exercise authority over coordination questions, compounds mediator failures across multi-hop operations, or defers the architectural question for vendor adjacencies is using a different composition, and the difference should be named.

Source paper

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235. Load-bearing sections for the present derivation: §2.3 (architectural distinctions from adjacent objects), §3.1 (path retraceability and adjacent-strand framing), §4.1 (governance boundary), §4.2 (AI-as-substrate-mediator role specification), §11.3 (substrate as source of truth as instantiated in the POC).

Related notes in this series

Li, W. (2026). *Mediator, Not Authority: The Architectural Role of LLMs in the Coordination Knowledge Substrate Pattern*. April 24, 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *Three Adjacencies: Why CKS Is Not RAG, Not Parametric Memory, and Not External Structured Memory*. April 24, 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *Three Positions, Not Four: Composing CKS Substrates with RAG Indexes, Fine-Tuned Models, Vector Databases, and External Knowledge Stores*. April 30, 2026. ORCID: 0009-0004-8065-3235.

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